



Wiremind

Hello app

Containerize and deploy a hello world app

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Agenda

01

INTRODUCTION

02

Main tasks

- Global architecture
- The application
- Containerization
- Kubernetes setup
- Beyond the Demo

03

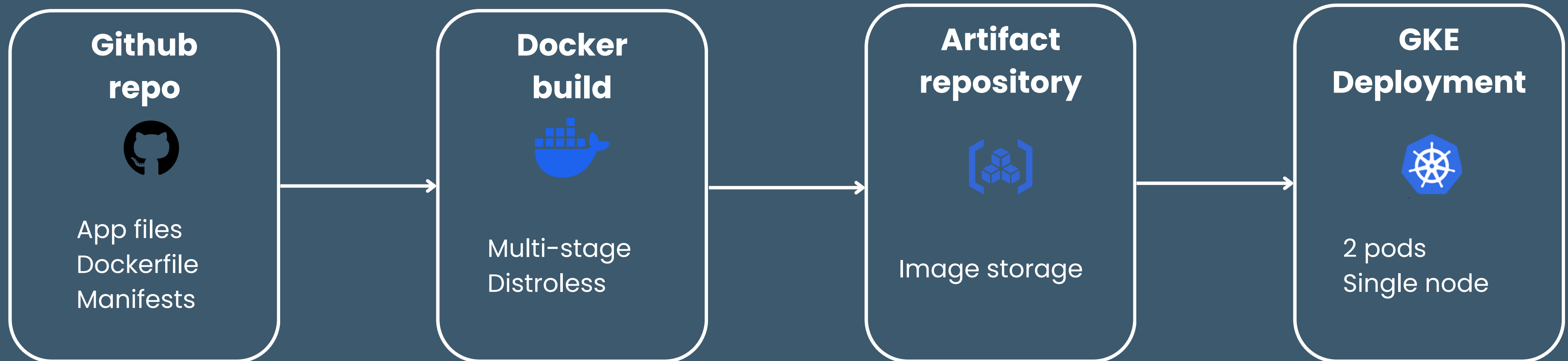
Demonstration



CONTEXT

- Build a simple "Hello" service with a Get name request
- Containerize and deploy it to GKE
- Explain the simplifications and tradeoffs
- Highlight what would be added in a production environment

HIGH-LEVEL ARCHITECTURE



CONTAINERIZATION

Best practices

- Multi-stage builds
- Small, minimal base image for production
- excludes files irrelevant to the build
- Pinned base image tags
- WORKDIR with absolute paths
- USER set to a non-root user rather than running as root

```
FROM golang:1.26-bookworm AS builder

WORKDIR /app

COPY app/go.* ./
RUN go mod download
COPY app/ .

RUN CGO_ENABLED=0 go build -trimpath -ldflags="-s -w" -o /out/hello-wiremind .

FROM gcr.io/distroless/static-debian13:nonroot

COPY --from=builder /out/hello-wiremind /app/hello-wiremind

USER nonroot:nonroot
EXPOSE 8080
ENTRYPOINT ["/app/hello-wiremind"]
```

*source: <https://docs.docker.com/build/building/best-practices/>

THE APPLICATION

GOLANG

- Static binary
- No runtime dependencies
- Small image footprint

2 ENDPOINTS

- GET /Hello
- GET /healthz

KUBERNETES SETUP

Deployment

- 2 Replicas
- LivenessProbe
- ReadinessProbe
- Resources management

Service

- Type load balancer

BEYOND THE DEMO

CI

Push to Repo

Automate container build,
test, and push to artifact
(Gitlab CI/Github actions)

Image pushed and tagged
to Artifact Registry

BEYOND THE DEMO

No manual deploy
(E.g: ArgoCD)

Handle observability
(metrics/logs)

DEMONSTRATION

Repo

`github.com/daadaamed/go-hello-app`

Service

`http://34.34.42.78/hello?firstname=Jane&lastname=Doe`

Health check

`http://34.34.42.78/healthz`

Questions welcome throughout.